CMPG 311
Group 12
Project PHASE 2
25 April 2025
Topic: Clinic Appointment System

GROUP MEMBERS

1.	Chauke DX	42705614
2.	Khoza S	32462409
3.	Mabasa NV	47337354
4.	Madonsela R	39803384
5.	Masinga S	44527802
6.	Mbombi RB	40676099
7.	Mukhumo B	38880199
8.	Munyai MG	42551900

1. CONCEPTUAL DESIGN

Business rules derived from clinic operations:

Patient Management Rules

- Each patient must have a unique Patient ID.
- A patient must provide full name, date of birth, contact details, and address upon registration.
- A patient's medical history, allergies, and insurance information must be stored securely.
- A patient can only view their own medical records through the secure patient portal.
- Patients can be placed on a waitlist if no appointment slots are available and will be notified when a slot opens.

Appointment Scheduling Rules

- Each appointment must have a unique Appointment ID.
- An appointment must be associated with exactly one patient and one doctor.
- Appointments must have a status (Scheduled, Completed, or Canceled).
- Appointments cannot be double-booked — no overlapping times for the same doctor or patient.
- Appointments can be booked, rescheduled, or canceled by patients or staff, depending on access rights.
- Doctors must define their availability in the system for appointments to be scheduled.
- System must automatically prevent scheduling outside a doctor's defined availability.
- Notifications must be sent (via SMS/email) for all booked, rescheduled, or canceled appointments.

Medical Staff Rules

- Each staff member (doctor, nurse) must have a unique Staff ID.
- Staff must be assigned a role (e.g., GP, Specialist, Nurse) and their contact and schedule must be stored.
- Doctors can only access and update consultation data for patients they are treating.
- Only medical staff can add/edit diagnosis, treatment plans, and prescriptions.

Medical Records Rules

- Each medical record must have a unique Record ID.
- Medical records must link to a specific patient and the doctor who handled the consultation.
- The system must support uploading/storing lab results, prescriptions, and treatment plans.
- Medical history and consultation notes must be timestamped and immutable (i.e., cannot be edited after finalization, unless by admin with logs).

Pharmacy and Inventory Rules

- Each medication or supply must have a unique Item ID.
- The system must track stock quantity, expiry date, and supplier info.
- Alerts must be generated when stock levels fall below the reorder level.
- Only pharmacy staff can update inventory records.

Billing & Payment Rules

- Each invoice must have a unique Invoice ID.
- Each invoice must be linked to one patient and must include total amount, payment method, and payment status.
- Invoices can be updated only by finance staff or administrators.
- Patients can view their own billing history but cannot modify payment records.

System Access and Security Rules

- All users must authenticate with a secure login based on their role: Admin, Doctor, Nurse, Receptionist, or Patient.
- Role-Based Access Control (RBAC) must be enforced — users can only access data and functions permitted to their role.
- All sensitive data (medical, financial, personal) must be encrypted in storage and during transmission.
- System must maintain audit logs for all data access, updates, and deletions.
- Regular data backups must be conducted and stored securely.

Operational and Performance Rules

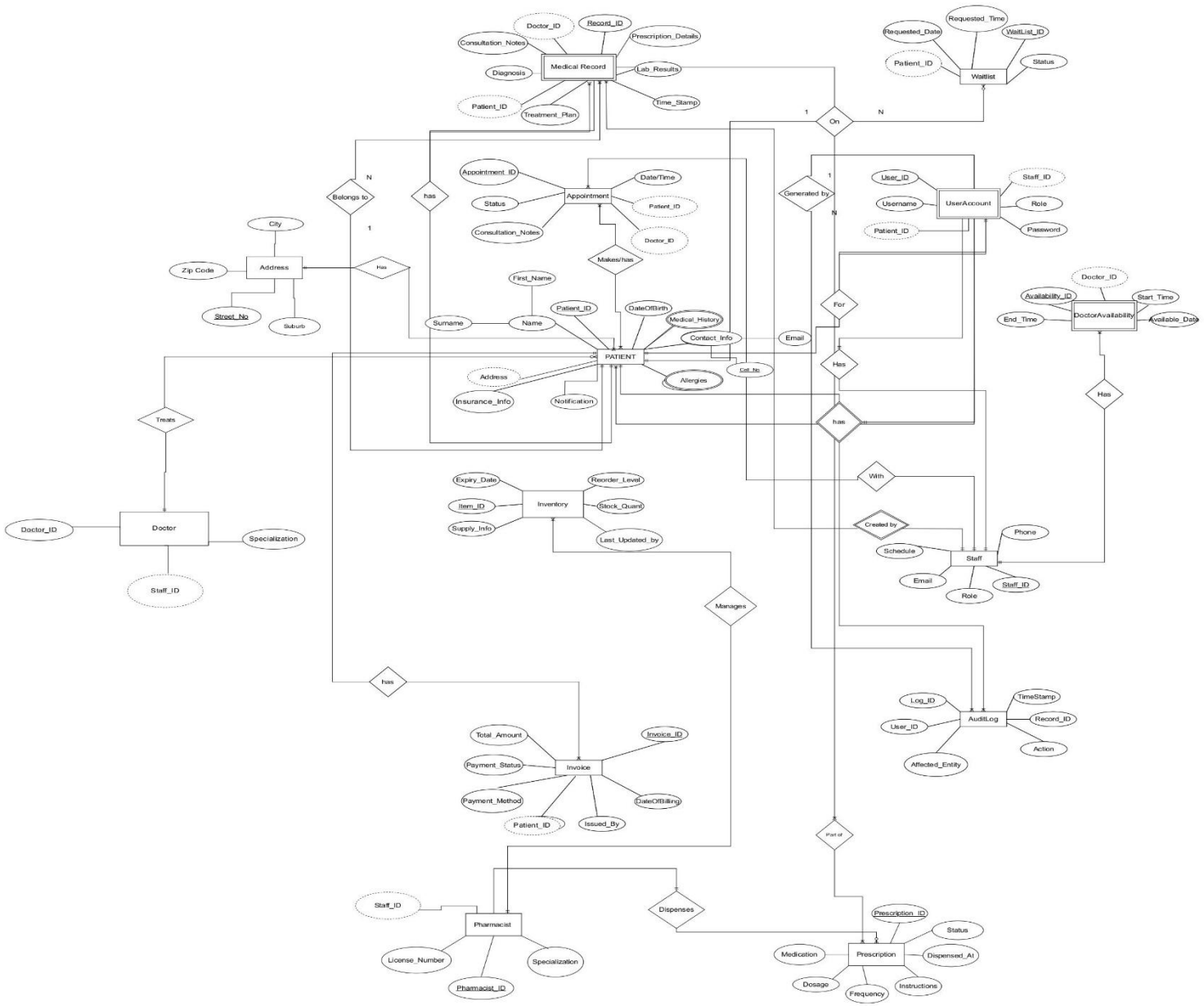
- The system must support at least 10,000 patient records initially, with scalability for future growth.
- All patient queries must return data in under 2 seconds for optimal user experience.

- The appointment system must prevent booking conflicts and detect changes in doctor availability in real-time.
- System updates (e.g., schedule changes) must not disrupt ongoing operations.

Compliance and Regulatory Rules

- All patient and staff data must comply with POPIA (Protection of Personal Information Act).
- Any data breach or unauthorized access attempt must be logged and reported to administrators immediately.

The Entity Relationship Diagram (ERD)



2. LOGICAL DATA MODEL

Using the conceptual model to map the logical data model:

i. Strong and weak entities

Strong entities	Weak entities
• Patient • Staff • Appointment • Invoice • Inventory • AuditLog • Waitlist • Prescription • Address	• DoctorAvailability • UserAccount • MedicalRecord

ii. Supertype/Subtype

SuperType	SubType
Staff	Doctor Pharmacist

The LOGICAL DATA MODEL:

1. **PATIENT** (Patient_ID (PK)INT, First_Name VARCHAR (50), Surname VARCHAR (50), DateOfBirth DATE, Cell_No VARCHAR (15), Email VARCHAR (100), Medical_History TEXT, Allergies TEXT, Contact_Info VARCHAR (255), Insurance_Info VARCHAR (255), Notification BOOLEAN)
2. **ADDRESS** (Address_ID(PK) INT, *PatientID(FK1)* INT, Street VARCHAR (100), Suburb VARCHAR (50), City VARCHAR (50), Zip_Code VARCHAR (10))
3. **APPOINTMENT** (Appointment_ID(PK) INT, *PatientID(FK1)* INT, *DoctorID(FK2)* INT, DateTime DATETIME, Consultation_Notes TEXT, Status VARCHAR (20))
4. **STAFF** (Staff_ID(PK) INT, Phone VARCHAR (15), Email VARCHAR (100), Schedule TEXT, Role VARCHAR (50))
5. **DOCTOR** (Doctor_ID(PK) INT, *Staff_ID* INT, Specialization VARCHAR (100))
6. **PHARMACIST** (Pharmacist_ID(PK) INT, *Staff_ID(FK1)* INT, License_Number VARCHAR (50), Specialization VARCHAR (100))
7. **MEDICAL_RECORD** (Record_ID(PK) INT, *Patient_ID(FK1)* INT, *Doctor_ID(FK2)* INT, Lab_Results TEXT, Treatment_Plan TEXT, Diagnosis TEXT, Consultation_Notes TEXT, Time_Stamp DATETIME)

8. **PRESCRIPTION** (Prescription_ID(PK) INT, *Record_ID(FK1)* INT, Medication VARCHAR (100), Dosage VARCHAR (50), Frequency VARCHAR (50), Instructions TEXT, Dispensed_At DATETIME, Status VARCHAR (20), Last_Updated_by INT)
9. **INVOICE** (Invoice_ID(PK) INT, *Patient_ID(FK1)* INT, Total_Amount DECIMAL, Payment_Method VARCHAR (50), Payment_Status VARCHAR (20), DateOfBilling DATE, Issued_By INT)
10. **INVENTORY** (Item_ID(PK) INT, *Stock_Quant(FK1)* INT, *Reorder_Level(FK2)* INT, Expiry_Date DATE, Supply_Info TEXT)
11. **USERACCOUNT** (User_ID(PK) INT, *Staff_ID(FK1*) INT, Username VARCHAR (50), Password VARCHAR (255))
12. **AUDITLOG** (Log_ID(PK) INT, *User_ID(FK1*) INT, Action VARCHAR (100), Affected_Entity VARCHAR (50), TimeStamp DATETIME)
13. **WAITLIST** (WaitList_ID(PK) INT, *Patient_ID(FK1)* INT, *Doctor_ID(FK2)* INT, Requested_Date DATE, Requested_Time TIME, Status VARCHAR (20))
14. **DOCTORAVAILABILITY** (Availability_ID(PK) INT, *Doctor_ID(FK1)* INT, Available_Date DATE, Start_Time TIME, End_Time TIME)

REFERENCES

1. Centre's for Disease Control and Prevention (CDC) (no date) *Best practices for telehealth*. Available at: https://www.cdc.gov/telehealth/best-practices.html (Accessed: 19 April 2025).
2. HPCSA (2023) *Ethical guidelines for healthcare practitioners*. Available at: https://www.hpcsa.co.za/Ethics/Guidelines (Accessed: 19 April 2025).
3. PostgreSQL Global Development Group, 2023. *PostgreSQL Documentation*. [online] Available at: https://www.postgresql.org/docs/ [Accessed 19 April 2025].
4. South African Medical Association (SAMA) (no date) *Practice management systems*. Available at: https://www.samedical.org/practice-management (Accessed: 19 April 2025).
5. World Health Organization, 2025. *Digital Health Guidelines (including EHRs and Telemedicine)*. [online] Available at: https://www.who.int/digital-health-guidelines [Accessed 19 April 2025].